

ANIRUDH PRAHLAD JOSHI

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Summary

Industry-focused Data Science Master's student at TU Hamburg with hands-on experience building and deploying machine learning solutions in both academic and industrial settings. Skilled in Python, PyTorch, SQL, and data engineering workflows, with applied experience in computer vision, NLP, and real-time analytics. Brings a strong engineering mindset, experience collaborating in agile teams, and a proven ability to translate complex data and AI models into scalable, business-ready solutions.

Education

Hamburg University of Technology (TUHH)

October 2024 – Present

Master of Science in Data Science, GPA: 2.6 out of 1.0

Hamburg, Germany

KLE TECHNOLOGICAL UNIVERSITY

August 2019 – June 2023

Bachelor of Engineering in Computer Science and Engineering, CGPA: 8.63 out of 10

Hubli, India

Professional Skills

Programming Languages: Python, C, C++

Machine Learning: Pytorch, NumPy, Pandas, scikit-learn, Deep Learning, Computer Vision, LLMs, RAG, LoRA

Database: SQL, MongoDB

Frameworks & Tools: LangChain, FAISS (vector database), Docker, Git

Cloud & Big data: Google Cloud Platform, AWS (EC2), Apache Kafka, Apache Flink

Spoken Languages: English (C1), German (A2), Kannada (native)

Professional Experience

Institute of Production Management and Technology, TUHH

February 2025 – Present

Research Assistant | VueJS, Python, MongoDB, Docker

Hamburg, Germany

- Developed a web application to digitalize the production process, designing responsive interfaces and integrating real-time data pipelines, resulting in faster production and improved workflow visibility for the management.
- Collaborated with backend developers and domain experts to translate operational requirements into user-friendly dashboards, enhancing system usability and data accessibility.
- Supported industrial analytics and automation initiatives by implementing data-driven features and process monitoring tools, contributing to improved decision-making efficiency.

Bosch Global Software Technologies

July 2023 – September 2024

Associate Software Engineer | C++, SQLite, Visual Studio, Qt, Git

Bangalore, India

- Worked on automotive infotainment navigation systems to handle location search and improve the performance, implementing optimized search algorithms and refining data indexing logic, leading to reduced query latency and improved user experience.

- Identified and resolved critical data inconsistencies and performance issues through systematic debugging and profiling, improving system reliability.
- Collaborated with software, hardware, and UX teams to validate features through integration, regression testing, on-device testing, ensuring stable releases.

Bosch Global Software Technologies

January 2023 – May 2023

Project Trainee | C++, SQLite, Visual Studio, Qt, Git

Bangalore, India

- Analyzed legacy map search modules to address high processing latency, refactored inefficient algorithms and optimized memory usage, achieving measurable reductions in response time.
- Conducted structured testing on embedded infotainment systems, validating performance improvements and deployment readiness.

Projects

Understanding Visual Encoders in Multi-Modal Language Models

PyTorch, Python, GPT-4.1, NumPy, Computer Vision

February 2025 – September 2025

- Investigated limitations in visual understanding of multimodal language models for VQA tasks by designing controlled experiments using custom datasets, enabling systematic evaluation of model behavior and weaknesses.
- Built a custom dataset from COCO images and GPT-4.1-generated question-answer pairs, increasing training data diversity and improving evaluation coverage.
- Trained and optimized BERT-BEiT-based vision-language models in PyTorch, achieving improved accuracy and more stable performance across VQA benchmarks.

Federated Sentiment Analysis

Apache Kafka, Apache Flink, JavaScript, Docker, Google Cloud Platform (GCP)

April 2025 – July 2025

- Designed a real-time pipeline to analyze large-scale social media sentiment data, integrating Kafka for ingestion and Flink for stream processing, enabling low-latency analytics.
- Implemented continuous model retraining using streaming data by integrating online learning workflows into the pipeline, improving sentiment classification accuracy and adaptability to evolving language patterns in real time.
- Containerized and deployed the system on GCP using Docker, ensuring scalability and fault tolerance.

Debiasing TextVQA

PyTorch, Python, NumPy, Computer Vision

January 2022 – July 2022

- Addressed dataset bias in TextVQA models, integrated VQA dataset as an external dataset and normalization techniques, improving fairness and generalization performance.
- Evaluated model robustness across diverse benchmarks, strengthening research validity.